

# Portfolio Optimisation through Random Matrix Theory

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(Dated: August 24, 2026)

Empirical correlation matrices estimated from a finite set of price observations are dominated by seemingly random noise, which in turn corrupts the covariance estimates which are relied heavily upon by mean-variance portfolio optimisers. To combat this, we can apply random matrix theory to such correlation matrices, using the Marchenko-Pastur law to separate eigenvalues associated to such randomness from informative eigenvalues, replacing them by their mean value. We then reconstruct a cleaned covariance matrix which we use to build a long only portfolio. On an out of sample test the denoised portfolio delivers a cumulative return of 26.83 % against 21.01 % for the portfolio built on the raw sample, a relative improvement of roughly 27.7 % (at time of testing).

## I. INTRODUCTION

Modern portfolio theory selects asset weights by trading expected return against risk, where risk is encoded in the covariance matrix of asset returns. In practice this matrix is not known and must be estimated from a finite sample of  $T$  historical observations for  $N$  assets. When  $N$  is comparable in size to  $T$ , the regime that arises for large equity groups such as the S&P 500, the sample covariance matrix is a poor estimate. It's smallest eigenvalues are underestimated, its largest overestimated, and the associated eigenvectors are heavily contaminated by noise. Because the Markowitz optimiser inverts this matrix, it amplifies the least reliable directions, producing unstable weights that perform poorly out of sample.

Random Matrix Theory provides a null model for this noise. If  $T$  independent and identically distributed Gaussian series of length  $N$  were purely random, the eigenvalue density of their correlation matrix would converge, in the limit  $N, T \rightarrow \infty$  with fixed ratio  $\gamma = N/T$ , to the Marchenko-Pastur distribution (7). Any eigenvalue that falls inside the MP support is therefore indistinguishable from noise, whereas eigenvalues lying above the upper edge  $\lambda_+$  carry genuine information. This observation motivates eigenvalue clipping: retain the signal eigenvalues, collapse the noisy bulk onto their mean, and rebuild a cleaned correlation matrix that is better conditioned than the raw estimate.

In this report we implement the full method on market data. We download five years of daily closing prices for the S&P 500 constituents, construct the empirical correlation matrix in a training window, denoise it using Marchenko-Pastur and compare the out-of-sample performance of maximum-Sharpe portfolios built from the raw and the cleaned covariance matrices.

## II. METHODS AND IMPLEMENTATION

The analysis is implemented in Python using `yfinance` for data acquisition, `pandas/numpy` for numerical linear algebra, and `scipy.optimize` for the constrained optimization.

### A. Data Construction

The company symbols are taken from the public S&P 500 listings and then 5 years of daily closing prices are downloaded using `yfinance`, forming a matrix  $P_{t,i}$ ,  $t$  indexing the time given by dates and  $i$  the companies. Then companies with significant incomplete data are filtered out, followed backfilling short gaps in prices for ease of analysis, resulting in  $P$  being roughly of size  $1254 \times 493$  at time of testing.

Prices are converted to logarithmic returns,

$$r_{t,i} = \ln\left(\frac{P_{t,i}}{P_{t-1,i}}\right), \quad (1)$$

which are used in preference to simple returns because they are time additive and symmetric, bringing the distribution closer to Gaussian, which is necessary to apply the following random matrix theory techniques. We write  $R_{t,i} = e^{r_{t,i}} - 1$  for the simple return when required. The returns are then split 80/20 for training and testing respectively.

### B. Empirical Correlation Matrix

Random matrix theory is formulated for standardized variables, so each asset return series is normalised (over training data),

$$x_{t,i} = \frac{r_{t,i} - \langle r_i \rangle}{\sigma_i}, \quad (2)$$

with  $\langle r_i \rangle$  and  $\sigma_i$  the sample mean and standard deviation of asset  $i$  over the training window. Collecting the standardized data in the  $T \times N$  matrix  $\mathbf{X}$ , the empirical correlation matrix is

$$\mathbf{E} = \frac{1}{T} \mathbf{X}^\top \mathbf{X}, \quad (3)$$

a symmetric positive semi-definite  $N \times N$  matrix. Its spectral decomposition,

$$\mathbf{E} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N), \quad (4)$$

is computed with `numpy.linalg.eigh`, yielding the eigenvalues  $\lambda_k$  and the orthonormal eigenvectors stored as the columns of  $\mathbf{V}$ .

### C. The Marchenko-Pastur Theorem

**Definition (Marchenko-Pastur Law).** Let  $\mathbf{X}$  be a  $p \times n$  matrix with entries  $\mathbf{X}_{ij}$  that are independent and identically distributed real random variables with mean  $\mathbb{E}[\mathbf{X}_{ij}] = 0$  and variance  $\mathbb{E}[\mathbf{X}_{ij}^2] = 1$ . Denote  $\Sigma$  the  $p \times p$  matrix is given by:

$$\Sigma = \frac{1}{n} \mathbf{X} \mathbf{X}^T \in \mathbb{R}^{p \times p}. \quad (5)$$

Let  $\Lambda_1, \Lambda_2, \dots, \Lambda_p$  denote the eigenvalues of  $\Sigma$ . We consider the empirical spectral measure of  $\Sigma$

$$\mu_p = \frac{1}{p} \sum_{j=1}^p \delta_{\Lambda_j}. \quad (6)$$

Supposing that all higher moments of  $\mathbf{X}_{i,j}$  are bounded, as  $p \rightarrow \infty$  and  $n \rightarrow \infty$  such that the aspect ratio  $p/n \rightarrow \gamma \in (0, 1]$ , the measure  $\mu_p$  converges almost surely to a deterministic distribution  $\mu$  with probability density function:

$$\frac{d\mu}{dx} = \begin{cases} \frac{1}{2\pi\gamma x} \sqrt{(a_+ - x)(x - a_-)} & \text{if } a_- \leq x \leq a_+ \\ 0 & \text{otherwise,} \end{cases} \quad (7)$$

where the upper and lower spectral bounds  $a_{\pm}$  are defined by:

$$a_{\pm} = (1 \pm \sqrt{\gamma})^2. \quad (8)$$

### D. Marchenko-Pastur Denoising

By comparing  $\mathbf{E}$  to the theoretical matrix  $\Sigma$  from the Marchenko-Pastur theorem, we establish that if the asset returns were purely uncorrelated random noise, all eigenvalues of  $\mathbf{E}$  would lie within the interval  $[\lambda_-, \lambda_+]$ , where:

$$\lambda_{\pm} = \left(1 \pm \sqrt{\frac{N}{T}}\right)^2. \quad (9)$$

Hence, any eigenvalue  $\lambda_i > \lambda_+$  of our empirical correlation matrix represents true signal. Conversely, all eigenvalues where  $\lambda_i \leq \lambda_+$  are classified as noise and are subject filtering (including those below  $\lambda_-$ ).

The noisy eigenvalues are replaced by their mean, preserving the trace  $\sum_k \lambda_k = N$ , which keeps the total variance fixed. The matrix is then reconstructed from the cleaned spectrum and the original eigenvectors,

$$\tilde{\mathbf{E}} = \mathbf{V} \Lambda^{\text{clean}} \mathbf{V}^T. \quad (10)$$

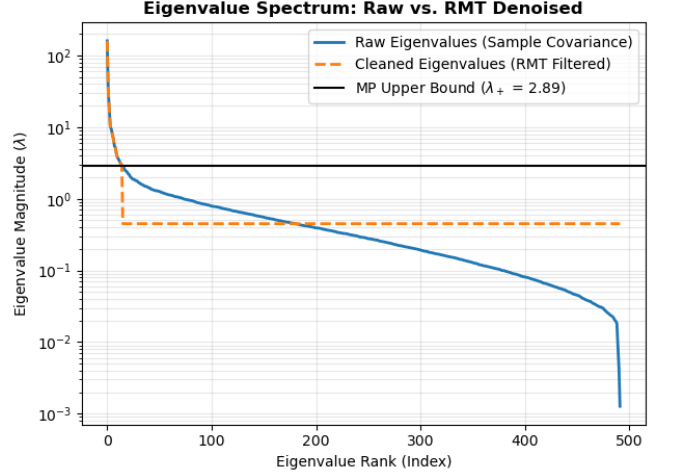


FIG. 1. Eigenvalue spectrum of the S&P 500 training correlation matrix, rank-ordered and plotted on a logarithmic scale. The raw sample eigenvalues are compared with the RMT-filtered spectrum. The horizontal line marks the Marchenko-Pastur upper edge  $\lambda_+$

After reconstructing the correlation matrix, renormalization is required because adjusting the eigenvalue spectrum alters the individual variances, causing diagonal elements to shift slightly from 1.0. Since a correlation matrix requires unit diagonal entries, we apply diagonal scaling,

$$\mathbf{E}_{ij}^{\text{clean}} = \frac{\tilde{\mathbf{E}}_{ij}}{\sqrt{\tilde{\mathbf{E}}_{ii} \tilde{\mathbf{E}}_{jj}}}, \quad (11)$$

to restore structure while retaining the denoised eigenspace. The effect of the filter on the spectrum is shown in Fig. 1.

### E. Portfolio optimization

Both correlation matrices are converted to annualized covariance matrices. The daily statistics are scaled to a 252 trading day year,

$$\mu_i = 252 \langle R_i \rangle, \quad s_i = \sqrt{252} \sigma(R_i), \quad (12)$$

and the covariance matrices are recovered using

$$\Sigma = \mathbf{D} \mathbf{C} \mathbf{D}, \quad (13)$$

where  $\mathbf{D} = \text{diag}(s_1, \dots, s_N)$ .

For a weight vector  $\mathbf{w}$  and risk-free rate  $r_f$ , the (annual) Sharpe ratio is

$$\mathcal{S}(\mathbf{w}) = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \Sigma \mathbf{w}}}. \quad (14)$$

TABLE I. Out-of-sample performance of the maximum-Sharpe portfolios built from the raw and RMT-denoised covariance matrices.

|                       | Raw     | RMT-cleaned |
|-----------------------|---------|-------------|
| Cumulative return $G$ | 21.01 % | 26.83 %     |
| Relative improvement  | —       | 27.70 %     |

The optimal long only, fully invested portfolio solves

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \mathcal{S}(\mathbf{w}) \quad \text{s.t.} \quad \sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq 1. \quad (15)$$

In code the maximization is converted to minimization of  $-\mathcal{S}$  and solved with Sequential Least-Squares Quadratic Programming (`scipy.optimize.minimize`, method `SLSQP`), starting from the equal-weight portfolio  $w_i = 1/N$ , with the budget constraint imposed as an equality and the no-short-selling condition as box bounds. Solving Eq. (15) with each covariance matrix yields  $\mathbf{w}_{\text{clean}}^*$  and  $\mathbf{w}_{\text{raw}}^*$ .

### F. Out-of-sample evaluation

The optimized weights are applied, unchanged, to the test window. The realized daily portfolio return is  $R_t = \mathbf{w}_i^{*\top} \mathbf{R}_{t,i}$ , from which the annualized out of sample Sharpe ratio is

$$\mathcal{S}_{\text{oos}} = \frac{\langle R \rangle - r_f/252}{\sigma_R} \sqrt{252}, \quad (16)$$

and the cumulative return over the test window is

$$G = \prod_{t \in \text{test}} (1 + R_t) - 1. \quad (17)$$

The relative improvement of the cleaned portfolio over the raw one is  $(G_{\text{clean}} - G_{\text{raw}})/|G_{\text{raw}}|$ .

Table I reports results and Fig. 2 shows the compounded growth of a unit investment across the test window for both portfolios.

## III. CONCLUSIONS

We have implemented a Random Matrix Theory denoising framework for portfolio construction on the S&P 500 companies. Filtering the empirical correlation matrix against the Marchenko-Pastur bounds, retaining signal eigenvalues above  $\lambda_+$  and collapsing the noise bulk onto its mean, producing a covariance estimate that is better conditioned and less contaminated by error than

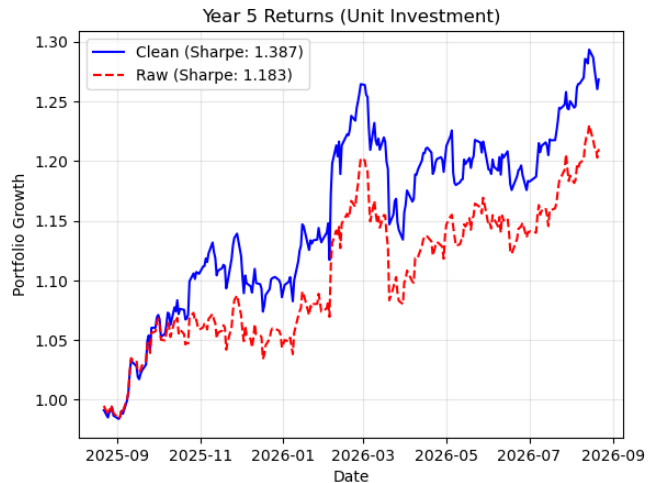


FIG. 2. Compounded growth of a unit investment over the out-of-sample test window for the RMT-cleaned portfolio and the raw-covariance portfolio.

the raw matrix. When fed to a long only maximum-Sharpe optimiser and tested on held-out data, the denoised portfolio achieved a cumulative return of 26.83 % versus 21.01 % for the raw-covariance portfolio, a relative gain of roughly 27.7 %.

The improvement is consistent with the established picture that the mean-variance optimiser is sensitive to noise in the small eigenvalues it inverts, and that suppressing that noise yields weights which generalise better out of sample. Furthermore, the gain is obtained without any additional return forecasting, it comes purely from a more faithful estimate of the risk structure. Thus confirming that spectral filtering is a low-cost and effective regulariser for high dimensional portfolio problems.

## IV. LIMITATIONS AND FUTURE WORK

*a. Expected-return estimation.* The optimiser uses the raw historical mean  $\boldsymbol{\mu}$  as the expected return, which is notoriously noisy. Shrinkage estimators, a Bayes/Black-Litterman prior, or a robust/return-agnostic objective would highlight the benefit of covariance cleaning from the instability of the mean.

*b. The denoising scheme* Flat eigenvalue clipping is the simplest RMT filter. More sophisticated alternatives such as the Ledoit-Wolf nonlinear shrinkage estimator or rotationally invariant estimators typically outperform clipping.

*c. Statistical modelling assumptions.* The Marchenko-Pastur law assumes independent and identically distributed Gaussian returns. Dropping assets with incomplete histories also introduces a bias toward long term firms. Using a longer or more accurate set of data at each point in time and validating the Gaussian assumption would make the null model hold

better.

*d. Evaluation robustness.* A single 80/20 chronological split gives one realisation of out of sample performance. Walk-forward or rolling-window backtesting across many periods would establish whether the observed edge is statistically significant. Reporting risk adjusted and drawdown metrics, and including transaction

costs and turnover, would make the comparison economically meaningful.

*e. Optimization and implementation.* The SLSQP solve starts from a equal-weight guess and could benefit from multiple restarts. Regularising the weights and testing the sensitivity of the result to the train/test ratio and to the risk-free rate would further show the robustness of the method.